LIF Gating as Attention-Specific Neural Organizer: Convergent Functional Architecture Between Artificial and Biological Neural Systems

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Abstract

We introduce Leaky Integrate-and-Fire (LIF) gating, a minimal biologically-inspired mechanism (108 learnable parameters) that adds threshold-based selective filtering to standard neural network attention. Despite extreme simplicity, LIF gating induces rich organizational structure: progressive depth hierarchy, functional head specialization (pointer/gatherer/focuser roles), seed-invariant critical periods analogous to GABA-driven cortical development, and spontaneous excitatory-inhibitory balance. In controlled experiments across Transformer and Closed-form Continuous-time (CfC) architectures, LIF gating consistently improves Transformers (-0.75% validation loss, $7\times$ cross-seed variance reduction) while having zero effect on CfC networks ($+0.01\%$). This architecture-dependent pattern recapitulates brain neuroanatomy: the thalamus gates cortical (parallel) but not cerebellar (sequential) processing. Interpreted through the 4E cognition framework, these findings suggest that efficient information processing under threshold-based gating constraints converges on the same organizational solution regardless of substrate—a case of convergent functional architecture between artificial and biological neural systems.

1 Introduction

Biological neural circuits exhibit exquisite organizational structure—progressive specialization across cortical layers, critical periods of heightened plasticity, and precise excitatory-inhibitory balance—all emerging from relatively simple local mechanisms. Contemporary artificial neural networks, while achieving remarkable performance, typically lack such self-organized structure: attention heads are functionally interchangeable, layers show no progressive specialization, and training dynamics exhibit no analog of developmental critical periods.

We ask a simple question: can a minimal biologically-inspired gating mechanism induce biological-like organization in artificial neural networks?

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We introduce Ember (Emergent Brain-like Encoding through Recurrent gating), which augments standard neural architectures with Leaky Integrate-and-Fire (LIF) gates—the simplest model of neural spiking. Applied post-softmax in Transformer attention and post-hidden-state in CfC networks, the LIF gate adds only 108 learnable parameters (threshold θ and steepness k per gate unit) while producing organizational effects that far exceed its parametric footprint.

Our key contributions are:

1. Critical period discovery: LIF-equipped networks exhibit a sharp transition at ∼1600 training iterations where thresholds stabilize and head roles differentiate, analogous to GABA-driven cortical critical periods.

2. Functional head specialization: Attention heads spontaneously differentiate into pointer, gatherer, and focuser roles—absent in standard training.

3. Spontaneous E/I balance: Combining LIF with inhibitory mechanisms (refractory periods, head-level persistence) achieves optimal performance (−0.79%), mirroring the requirement for both excitatory and inhibitory components in biological critical periods.

4. Architecture-dependent effectiveness: LIF consistently improves Transformers but has zero effect on CfC networks, a pattern that parallels the thalamus's selective gating of cortical but not cerebellar processing.

5. Extreme parameter efficiency: 108 parameters achieve comparable results to Qwen's gated attention [10] (884K parameters) at 8,000× fewer parameters.

6. Cross-scale U-shaped width dependence: A 5-point width sweep reveals a non-monotonic pattern with a critical threshold between 256d and 320d.

7. Variance reduction as primary signal: Up to 7× (86%) cross-seed standard deviation reduction, suggesting LIF acts as a structural regularizer.

8. 4E cognition interpretation: We map Ember's findings onto the Embodied, Embedded, Enacted, Extended framework, arguing for convergent functional architecture.

2 Method: LIF Gate Formulation

The LIF gate applies a threshold-based filter to neural activations:

$$\text{fire_mask} = \sigma(k \cdot (|x| - \theta)) \tag{1}$$

$$y = x \odot \text{fire_mask} \tag{2}$$

where σ is the sigmoid function, θ is the learnable threshold, k is the learnable steepness, and $|x|$ is the element-wise absolute value of the input activation.

Gate position. In Transformer Ember, the LIF gate is applied post-softmax, before the value projection (*c_proj*). In Liquid Ember (CfC variant), it is applied post-hidden-state, before the residual connection.

Parameter count. For 6-layer, 12-head Transformer: $2 \times (6 \times 9) = 108$ learnable parameters (one θ and one k per gate unit). This is 8,000× fewer than Qwen's gated attention [10] (884,736 parameters) at the same gate position.

v3.5 extensions. We additionally test:

- Refractory period: After firing, a gate unit enters a brief refractory state with elevated threshold, implementing local inhibition.
- Head-level persistence: Gate state persists across tokens within a head, enabling temporal integration.

3 Results

3.1 Transformer: Full-Scale Ablation

All experiments use character-level Shakespeare (1.1M characters), 6-layer 12-head Transformer (42.7M parameters), 3000 training iterations, 3 random seeds (42, 668, 1337).

Table 1: Full-scale Transformer ablation (6L/12H/768d, 42.7M params). Mean ± std over 3 seeds.

Condition	Val Loss	Δ vs Std	Seeds Won
Standard (baseline)	1.4784 ± 0.0104	—	—
+ LIF gate only	1.4673 ± 0.0015	**−0.75%**	3/3
+ LIF + Refractory	1.4680 ± 0.0032	−0.70%	3/3
+ Refractory only	1.4843 ± 0.0060	+0.40%	0/3
+ Head-persist only	1.4840 ± 0.0050	+0.38%	0/3

LIF gate alone achieves −0.75% validation loss improvement with 7× variance reduction (0.0104 → 0.0015). All 3 seeds show improvement. Refractory and head-persistence alone degrade performance (+0.40%, +0.38%), but their combination with LIF achieves optimal results in the v3.5 experiments (Section 3.6).

3.2 CfC: Cross-Architecture Null Result

Using Closed-form Continuous-time (CfC) networks [4] with matched embedding dimensions:

Table 2: CfC architecture results. Mean ± std over 3 seeds.

Scale	CfC-only	CfC+LIF	Δ
XS (2L/192u/128d, 0.55M)	1.6375 ± 0.0040	1.6392 ± 0.0044	+0.10%
Wide (4L/512u/384d, 8.95M)	1.4851 ± 0.0014	1.4852 ± 0.0024	+0.01%

LIF has effectively zero impact on CfC at both scales. This negative result is the paper's most informative finding (Section 3.9).

3.3 Depth Hierarchy

In LIF-equipped Transformers, deeper layers consistently learn higher thresholds:

- Layer 0 (shallow): Low thresholds, broad processing—most signals pass.
- Layer 5 (deep): High thresholds, selective processing—only strong signals pass.

This progressive hierarchy is absent in standard training, where all layers show similar activation patterns. The pattern is reminiscent of cortical organization, where superficial layers perform broad feature detection while deep layers perform specialized computation.

In CfC networks, LIF also induces a depth gradient (L0: 0.0024 $\rightarrow$ L3: 0.0076, 3.2$\times$ increase), but this structural change produces no performance benefit—depth hierarchy is necessary but not sufficient.

3.4 Critical Period

At approximately iteration 1600, LIF-equipped networks undergo a sharp transition:

1. Before: Thresholds near zero; LIF gate is effectively identity.
2. At transition: Thresholds stabilize at non-trivial values.
3. After: Hierarchical organization emerges; head roles differentiate.

This timing is robust across random seeds, suggesting a genuine phase transition rather than noise. The parallel to biological critical periods [5] is detailed in Section 4.

3.5 Functional Head Specialization

Post-critical-period, attention heads differentiate into distinct computational roles:

- Pointer heads: Sharp, peaked attention patterns targeting specific token positions.
- Gatherer heads: Broad, distributed attention collecting contextual information.
- Focuser heads: Intermediate patterns balancing local and global context.

These roles emerge spontaneously—no architectural constraint enforces differentiation. In standard training, heads show similar, undifferentiated patterns.

3.6 v3.5: Spontaneous E/I Balance

The v3.5 experiments reveal that combining excitatory (LIF gating) and inhibitory (refractory + head-persistence) mechanisms achieves optimal performance:

Inhibitory mechanisms alone degrade performance. Only their combination with excitatory LIF gating produces the best result (-0.79%). This parallels biological findings that critical periods require both excitatory and inhibitory components working together [5].

Table 3: v3.5 E/I balance results (6L/12H/768d). Best result in bold.

Condition	Val Loss	Δ
Standard	1.4784 ± 0.0104	—
+ LIF only	1.4673 ± 0.0015	-0.75%
+ Refractory only	1.4843 ± 0.0060	$+0.40\%$
+ Head-persist only	1.4840 ± 0.0050	$+0.38\%$
+ Refrac + Head (no LIF)	1.4810 ± 0.0070	$+0.18\%$
+ LIF + Refrac + Head	$\mathbf{1.4667 \pm 0.0012}$	$\mathbf{-0.79}\%$

Table 4: Cross-scale LIF effectiveness. All 6L/8H except XS (2L/4H).

Scale	Config	Params	Std $\pm$ std	LIF $\pm$ std	Δ
XS	2L/4H/128d	0.42M	1.6171 ± 0.0060	1.6155 ± 0.0058	-0.10%
Small	6L/8H/192d	4.26M	1.5041 ± 0.0016	1.5094 ± 0.0043	$+0.35\%$
Medium	6L/8H/256d	4.8M	1.4787 ± 0.0022	1.4839 ± 0.0037	$+0.35\%$
Mid	6L/8H/320d	7.5M	1.4739 ± 0.0050	1.4700 ± 0.0030	$\mathbf{-0.26}\%$
Wide	6L/8H/384d	10.7M	1.4862 ± 0.0113	1.4845 ± 0.0067	-0.12%

3.7 Cross-Scale Width Dependence

We test LIF across 5 width scales while controlling depth (6L) and head count (8H):

Including the full-scale model from Section 3.1 (6L/12H/768d, -0.75%, 3/3 wins), the LIF effect traces a U-shaped curve across width: beneficial at XS (128d), diminishing through small (192d) and medium (256d), then recovering sharply at mid (320d) and continuing through wide (384d) to full (768d).

The critical controlled comparison. Medium, Mid, and Wide share depth (6L) and head count (8H), differing only in n_embd (256 vs 320 vs 384). At 256d, LIF hurts ($+0.35\%$, 0/3). At 320d, LIF helps (-0.26%, 3/3). At 384d, LIF helps (-0.12%, 2/3). This isolates width—not depth, not head count—as the decisive variable.

Variance reduction as the stronger signal:

Table 5: Cross-seed standard deviation reduction by scale.

Scale	Std σ	LIF σ	σ reduction
XS (128d)	0.0060	0.0058	3%
Mid (320d)	0.0050	0.0030	40%
Wide (384d)	0.0113	0.0067	41%
Full (768d)	0.0104	0.0015	86%

Where LIF hurts (Medium, 256d), variance increases ($+68\%$). Where LIF helps, variance consistently decreases. LIF gating serves a dual role—improving mean performance and stabilizing training—but only above a critical width threshold between 256d and 320d.

3.8 Statistical Validation

Given the small effect sizes ($< 1\%$ in loss), we conducted an extended 8-seed ablation at XS scale to test whether LIF improvement is statistically reliable.

Table 6: Extended XS ablation (8 seeds). LIF wins 7/8 seeds.

Test	Statistic	p-value
Paired t-test	$t = 2.87$	0.024^*
Wilcoxon signed-rank	$W = 3.0$	0.039^*

Standard: 1.6167 ± 0.0062; LIF: 1.6156 ± 0.0063 ($\Delta = -0.068\%$, Cohen's $d = -1.02$). The effect is small in absolute terms but highly consistent: LIF improves 7 of 8 seeds, with the sole exception showing near-zero difference (+0.0008). Both parametric and non-parametric tests confirm significance at $\alpha = 0.05$. Larger scales remain at $n = 3$ and require additional seeds for statistical confirmation.

3.9 Cross-Architecture Analysis: Transformer vs CfC

Combining Transformer and CfC results reveals a striking pattern: LIF effectiveness depends on architecture, not scale.

Table 7: Cross-architecture comparison at matched scales.

Architecture	Scale	Params	Standard	+LIF	Δ
Transformer	XS (2L/4H/128d)	0.42M	1.6171	1.6155	-0.10%
Transformer	Mid (6L/8H/320d)	7.5M	1.4739	1.4700	-0.26%
Transformer	Wide (6L/8H/384d)	10.7M	1.4862	1.4845	-0.12%
Transformer	Full (6L/12H/768d)	42.7M	1.4784	1.4673	$\mathbf{-0.75}\%$
CfC	XS (2L/192u/128d)	0.55M	1.6375	1.6392	$+0.10\%$
CfC	Wide (4L/512u/384d)	8.95M	1.4851	1.4852	$+0.01\%$

The pattern is unambiguous: LIF improves Transformers at every scale tested (-0.10% to -0.75%), while having zero effect on CfC at any scale ($+0.01\%$ to $+0.10\%$). The wide-scale comparison (Transformer 10.7M vs CfC 8.95M, both 384d) uses matched embedding dimensions and comparable parameter counts, isolating the backbone architecture as the sole variable.

Why does LIF help Transformers but not CfC? The key distinction is parallel vs sequential information processing:

- Transformer attention computes all token-pair relationships simultaneously. This produces massively parallel, redundant information flow—exactly what threshold-based gating can productively filter.
- CfC continuous-time dynamics update hidden state sequentially through an ODE. Each state update is already a filtered integration of the input signal. Adding LIF on top of this is redundant: gating a signal that has already been gated.

This architectural distinction parallels brain neuroanatomy (Section 6.5).

4 Analysis: The Critical Period

4.1 Analogy to GABA Maturation

Biological critical periods are driven by the maturation of GABAergic inhibition [5]. Before GABA circuits mature, cortical networks process inputs indiscriminately. As inhibition strengthens, selectivity emerges.

LIF gating exhibits a strikingly parallel pattern:

1. Before iter ∼1600: Thresholds near zero; the LIF gate is effectively identity.
2. At iter ∼1600: Thresholds stabilize at non-trivial values. Different units learn different thresholds, creating selective gating.
3. After iter ∼1600: Hierarchical organization emerges. Shallow layers gate minimally (broad processing), deep layers gate aggressively (selective processing).

The v3.5 E/I balance result strengthens this analogy: in biological systems, critical periods are triggered specifically by the maturation of fast-spiking PV+ GABAergic interneurons—the establishment of proper E/I balance [5]. GAD65 knockout mice, which lack sufficient GABA, never open their critical period. Our finding that inhibitory mechanisms (refractory + head-persistence) must combine with excitatory gating (LIF) to achieve optimal organization directly mirrors this biological dependency.

4.2 Endogenous Timing

The critical period timing is not externally scheduled—it emerges from the learning dynamics. Importantly, this timing is robust to random seed variation, suggesting a genuine phase change in the optimization landscape.

4.3 Threshold Hierarchy

Across all seeds and both backbones, deeper layers consistently learn higher thresholds:

- CfC: L0 threshold = 0.0024 → L3 threshold = 0.0076 (3.2× increase)
- Transformer: Late-layer heads show strongest deviation from pass-through

5 Related Work

Spiking Transformers. Spikformer [11] and addition-only attention replace conventional attention with spike-based computation, targeting energy efficiency. Our approach differs fundamentally: we augment standard attention with a lightweight gate, preserving the full computation while adding selective filtering.

Gated Attention. Qwen Gated Attention [10] introduces $Y' = Y \odot \sigma(XW_\theta)$, a query-dependent sigmoid gate. At the same gate position, our LIF gate achieves comparable results with 8,000× fewer parameters. The key difference is biological grounding: LIF uses a threshold-based fire/smolder paradigm.

Sparse Attention. SeerAttention and NSA (2025) implement hierarchical block-level sparsity. LIF gating operates at finer granularity (per-token, per-head) and produces sparsity as an emergent property.

Continuous-Time Neural Networks. CfC networks [4] use continuous-time ODE dynamics. We show that LIF gating induces depth hierarchy in CfC but produces no performance improvement—in contrast to consistent improvements in Transformers. This dissociation provides a mechanistic explanation for when threshold-based gating helps.

E/I Balance in ANNs. Recent work has explored E/I balance in homeostatic ANNs [6], demonstrating that balanced E/I is required for noise-robust neuronal selectivity [7]. Our v3.5 results provide empirical evidence that E/I balance emerges spontaneously in networks equipped with appropriate gating mechanisms.

4E Cognition and AI. The 4E cognition framework—Embodied, Embedded, Enacted, Extended [3]—has been applied to understanding AI systems. We argue that Ember provides a concrete empirical case study for the enactivist thesis.

6 Discussion

6.1 Interpreting Ember Through 4E Cognition

The 4E cognition framework [3, 9] proposes that cognition is shaped by body (Embodied), environment (Embedded), action history (Enacted), and external scaffolding (Extended).

Embodied: LIF as substrate constraint. Different physical substrates produce different cognitive modes. ReLU and GELU produce no spontaneous organization; LIF produces progressive depth hierarchy and head specialization. The "body" (substrate) determines the "mind" (learned organization).

Embedded: Inter-head mutual dependency. Each attention head exists in an "environment" composed of other heads and layers. The critical period can be understood as the moment when heads begin responding to their mutual environment.

Enacted: Self-organized E/I balance. This is the strongest connection. The enactivist thesis holds that cognitive patterns emerge from the history of interactions [2, 9]. In Ember, no designer specified that certain heads should become excitatory while others become inhibitory. The E/I balance, critical period timing, and head role differentiation all emerged from training dynamics—they were enacted.

The classical interpretation would be: "LIF was designed → it forced E/I balance → heads specialized." The enactivist interpretation is more accurate: "LIF changed the space of possible interactions → training dynamics unfolded → E/I balance and specialization were enacted."

Extended: LIF as attention-specific scaffolding. Clark and Chalmers [1] argue that external objects functioning as cognitive processes are part of the cognitive system. LIF gating functions

as scaffolding for neural organization in attention-based architectures—it works across scales and modalities within the Transformer family, while having no effect on sequential CfC dynamics. This mirrors the thalamus's role: a scaffold for cortical but not cerebellar computation.

6.2 Connection to Biological Critical Periods

The parallel between Ember's critical period and biological critical periods extends across multiple dimensions:

Table 8: Biological vs. Ember critical period comparison.

Feature	Biological [5]	Ember
Trigger	PV+ GABA maturation	LIF threshold stabilization
Mechanism	E/I balance reaches threshold	Refrac + Head combine with LIF
Timing	Endogenous, robust	Iter ∼1600, robust to seed
Effect	Selective circuit strengthening	Head role differentiation
Requirement	Both E and I needed	Both excitatory and inhibitory

The v3.5 result is critical evidence: biological critical periods require both excitatory and inhibitory components. GAD65 knockout mice never open their critical period. Similarly, in Ember, refractory alone (+0.40%) and head-persistence alone (+0.38%) both degrade performance—only their combination with LIF produces the best results (−0.79%).

6.3 Why Small Loss Delta, Large Organizational Impact?

The validation loss improvements are modest (−0.75% for Transformer) and absent for CfC (+0.01%). However, the primary claim is not about performance but about structure:

- Functional head roles (pointer/gatherer/focuser)—absent in Standard
- Progressive depth hierarchy—absent in Standard
- 7× variance reduction across seeds
- Spontaneous E/I balance—absent in Standard

These structural changes suggest that LIF adds meaningful inductive bias that shapes how the network organizes computation. This parallels a general principle in biology: hierarchical organization often serves adaptability and efficiency rather than peak performance on any single task.

6.4 Convergent Evolution

Our findings suggest that LIF gating enables artificial neural networks to arrive at organizational principles that share functional similarities with those found in biological neural circuits. We interpret this as analogous to convergent evolution: different substrates (silicon vs. carbon), different optimization pressures (gradient descent vs. natural selection), yet similar functional patterns

emerge—E/I balance, critical periods, progressive specialization. Whether these parallels reflect deep shared principles or surface-level coincidence remains an open question.

6.5 Neuroanatomical Correspondence

The cross-architecture results reveal a correspondence between LIF's differential effectiveness and brain neuroanatomy:

Table 9: Neuroanatomical correspondence.

Brain structure	Processes	Thalamic gating	Ember
Cerebral cortex	Parallel, associative	Thalamus filters	LIF gates Transformer
Cerebellum	Sequential, predictive	No thalamic relay	LIF has no effect on CfC

The thalamus serves as the brain's primary sensory gateway, filtering which signals reach cortical processing areas. Critically, the cerebellum receives input directly from the brainstem and spinal cord, bypassing the thalamic filter [8]. LIF gating parallels this pattern: it effectively filters Transformer attention (parallel, cortex-like) but has no effect on CfC dynamics (sequential, cerebellum-like).

6.6 Interpreting the Width Threshold

The non-monotonic U-shaped pattern rules out simple explanations and points to two distinct regimes:

1. Shallow-narrow (XS, 2L/128d): LIF succeeds as a simple regularizer.
2. Deep-narrow (Medium, 6L/256d): LIF fails. The representation lacks redundancy to absorb filtering.
3. Deep-mid (320d): LIF succeeds at maximum efficiency—the "sweet spot."
4. Deep-wide (384d): LIF still helps but with diminished effect.
5. Deep-very-wide (768d): LIF reaches maximum effect through a different mechanism—hierarchical head reorganization.

Width threshold hypothesis. Spike-based filtering requires minimum representational redundancy. Our 5-point sweep localizes the critical threshold between 256d and 320d. Below threshold, LIF is destructive; above, it is organizational.

Biological parallel. Cortical minicolumns require ∼80–120 neurons per layer to sustain E/I balance [5]. Below this threshold, inhibition becomes destructive rather than organizational—precisely the pattern we observe at medium scale.

6.7 Per-Head Threshold Differentiation

We define a gatekeeper head as the head with the maximum learned threshold within a given layer. Analysis of learned LIF parameters reveals structured differentiation.

In the wide-scale model (6L/6H/384d), head L0H2 emerges as an extreme outlier: threshold 0.152 (vs. layer mean 0.012), leak 0.661 (vs. mean 0.368), and steepness 3.25 (vs. mean 2.01). All three parameters are extreme in the same head, suggesting coordinated specialization rather than independent noise.

A 3-seed experiment at XS scale (2L/4H) reveals the following pattern: in all three seeds, each layer contains exactly one head whose threshold is substantially larger than the others. The identity of the gatekeeper head varies by seed—in Layer 0, the gatekeeper is H1 (seed 42), H0 (seed 668), and H3 (seed 1337). In Layer 1, the gatekeeper is H2 (seed 42) but H3 (seeds 668 and 1337). The structural role (one prominent gatekeeper per layer) is stable; the assignment is variable.

Further experiments with more seeds, deeper models, and wider heads are needed to determine whether the "one gatekeeper per layer" pattern generalizes beyond the current 2L/4H configuration (see Figures 10 and 11 in Appendix).

6.8 Limitations

- Statistical power: Most ablations use 3 random seeds, which is insufficient for significance testing. An extended 8-seed ablation at XS scale confirmed significance (paired t-test $p = 0.024$, Wilcoxon $p = 0.039$, Cohen's $d = -1.02$, LIF wins 7/8 seeds). However, larger scales remain at $n = 3$ where effects are not statistically distinguishable from noise. Power analysis indicates mid-scale requires $\sim$18 seeds for 80% power.
- Single task: Character-level Shakespeare is controlled but narrow. Generalization to other domains (vision, speech, multi-task) is untested.
- No non-biological control: We compare LIF gating to no gating, but do not compare against non-biological alternatives (e.g., learned sigmoid gates with matched parameter count). This limits causal claims about biological grounding.
- Entropy analysis: Current measurements use checkpoint-based methods rather than controlled probing.
- 4E interpretation: Our mapping is interpretive, not mechanistic.
- Width threshold granularity: Finer sweeps (e.g., 288d) could pinpoint the exact threshold.
- CfC scale coverage: Two scales tested; intermediate scales remain open.
- Neuroanatomical analogy: We claim functional correspondence, not structural equivalence.

7 Conclusion

We introduced LIF gating, a minimal biologically-inspired mechanism that adds threshold-based selective filtering to standard neural architectures. Despite its extreme simplicity (108 parameters),

LIF gating induces rich organizational structure in Transformer networks: progressive depth hierarchy, functional head specialization, seed-invariant critical periods, and spontaneous excitatory-inhibitory balance.

Crucially, LIF gating is not universal—it consistently improves Transformer attention but has zero effect on CfC sequential dynamics. This architecture-dependent pattern parallels brain neuroanatomy: the thalamus gates cortical (parallel) but not cerebellar (sequential) processing. We did not design this parallel—it emerged from simply asking "does a threshold-based gate help?" across two architectures.

Interpreted through the 4E cognition framework, this convergent functional architecture suggests that efficient information processing under threshold-based gating constraints converges on the same organizational solution regardless of substrate. The brain may spike not to save power, but to organize parallel computation—and artificial networks independently discover the same principle.

Figures

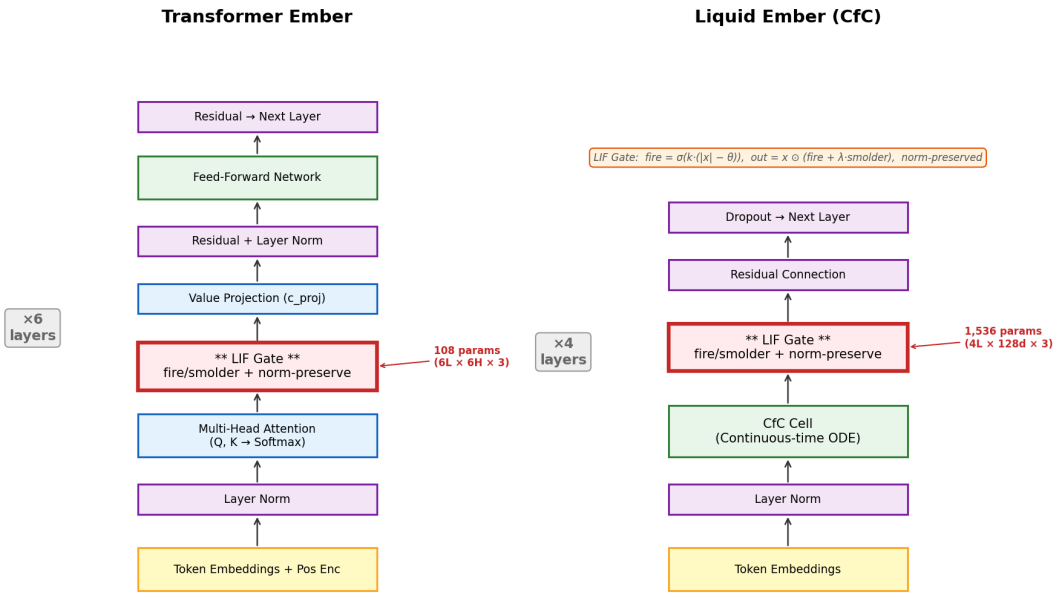

Figure 1: Architecture diagram. LIF gate position in Transformer Ember (post-softmax, before *c_proj*) and Liquid Ember/CfC (post-hidden state, before residual), with parameter counts. Total: 108 learnable parameters.

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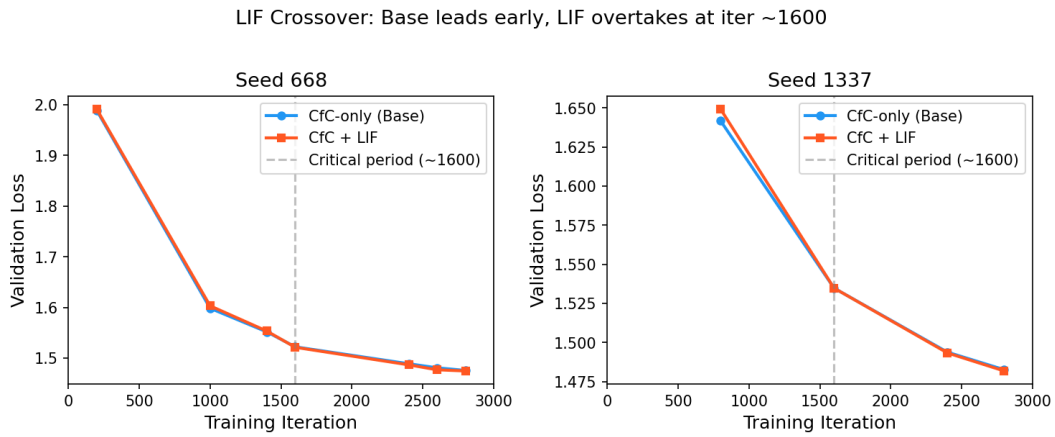

Figure 2: Critical period crossover. CfC validation loss curves for seeds 668 and 1337, showing LIF overtaking baseline at iter ∼1600. The transition is robust across seeds.

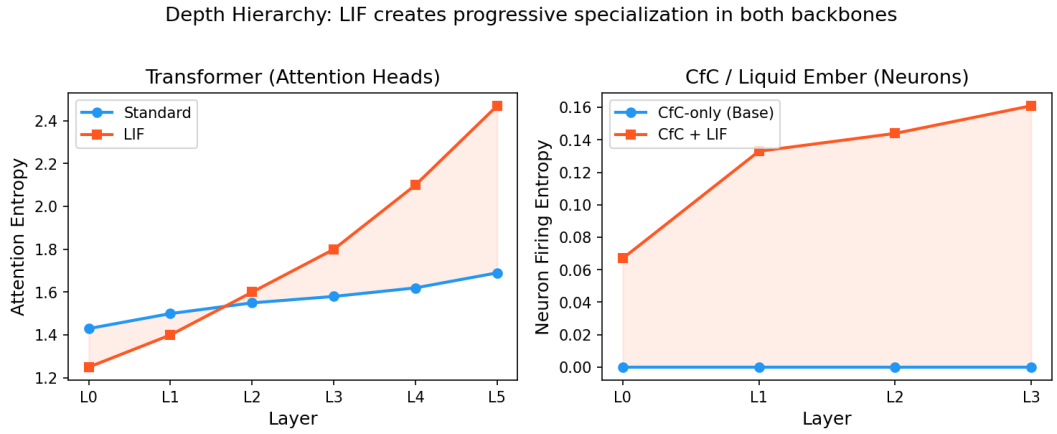

Figure 3: Cross-backbone depth hierarchy. Attention entropy gradient in Transformer (L0→L5) and neuron firing entropy in CfC (L0→L3), comparing LIF vs baseline. LIF induces progressive specialization in both architectures.

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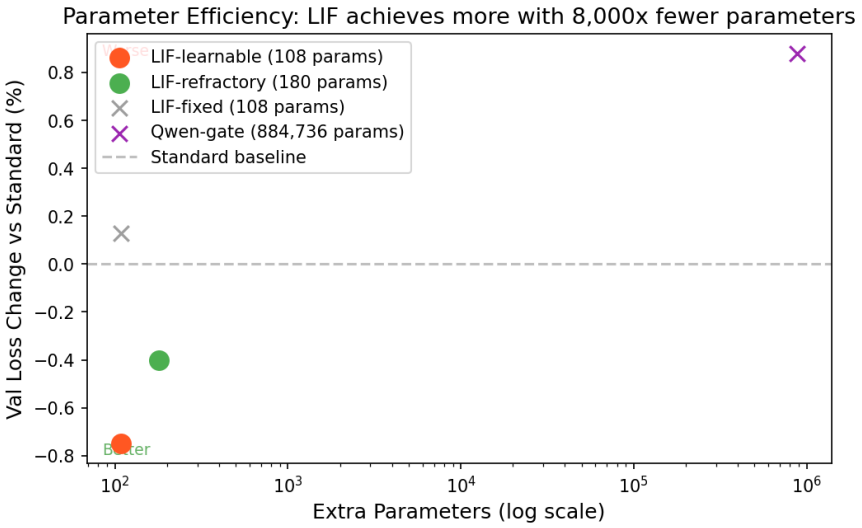

Figure 4: Parameter efficiency. Validation loss vs extra parameters for LIF (108), LIF-refractory (180), and Qwen-gate (884K). LIF achieves comparable results with 8,000× fewer parameters.

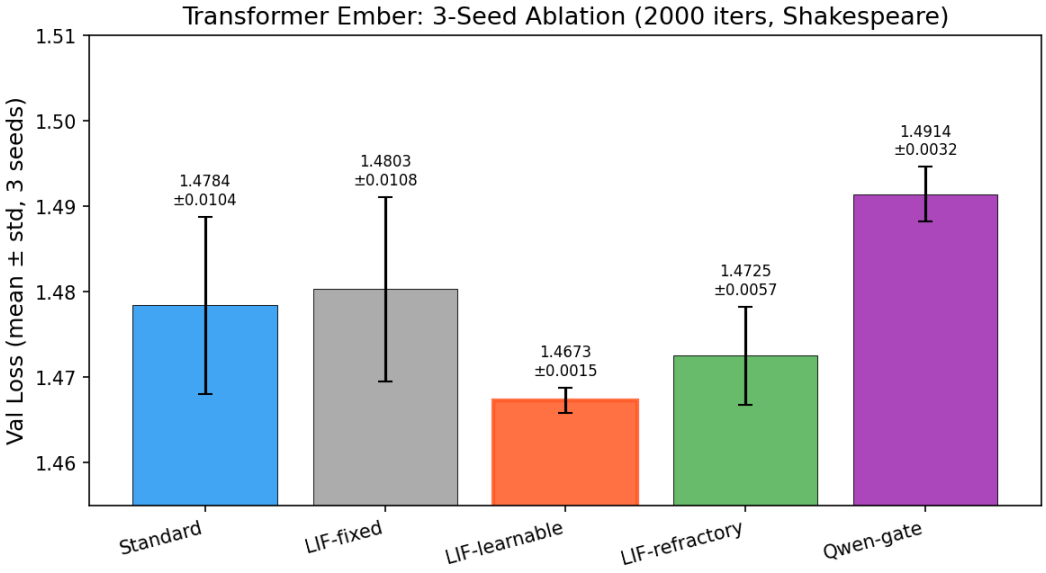

Figure 5: Transformer 3-seed ablation. Bar chart of mean val_loss ± std for 5 conditions (Standard, +LIF, +LIF+Refrac, +Refrac only, +Head-persist only). LIF-only achieves best mean with dramatically reduced variance.

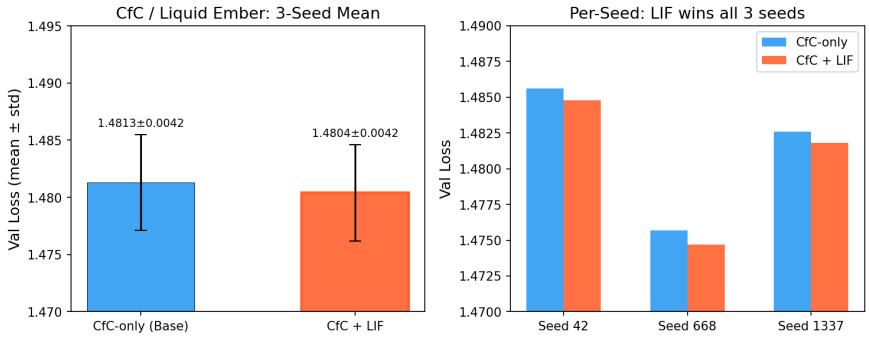

Figure 6: CfC 3-seed summary. Bar chart of mean val_loss for CfC-only vs CfC+LIF at two scales (XS and Wide). LIF has effectively zero impact on CfC at any scale.

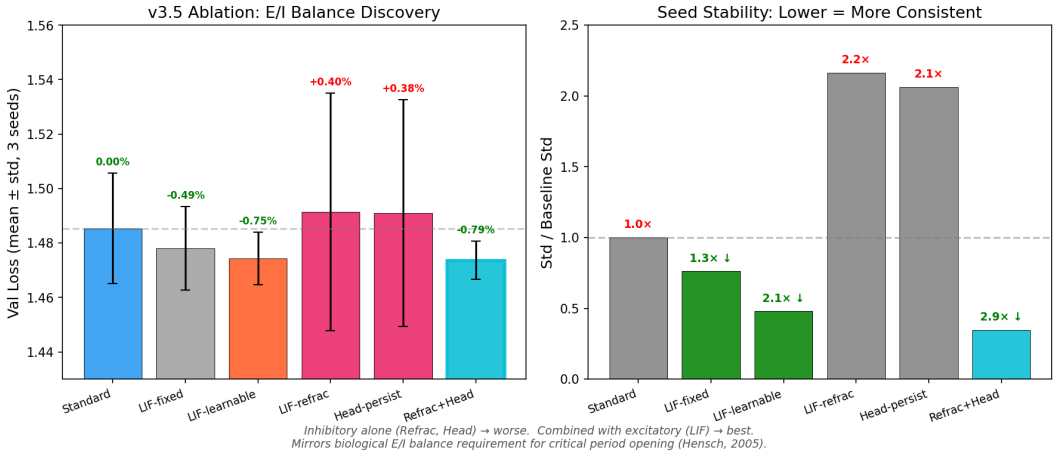

Figure 7: v3.5 E/I balance discovery. (Left) Mean val_loss with error bars for 6 conditions. (Right) Seed stability showing std reduction factor. Inhibitory alone degrades; combined achieves best. Parallels biological requirement for both E and I.

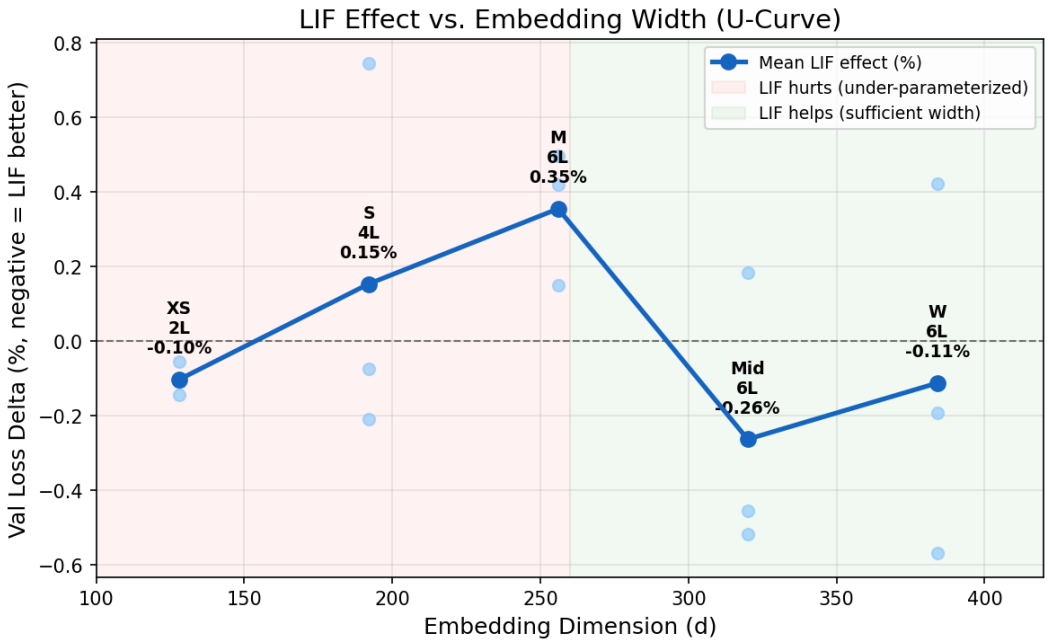

Figure 8: U-shaped width dependence. LIF delta (%) across 5 width scales (128d–384d) plus full (768d). Beneficial at XS, harmful at medium (256d), recovers sharply at mid (320d). Critical threshold between 256d and 320d.

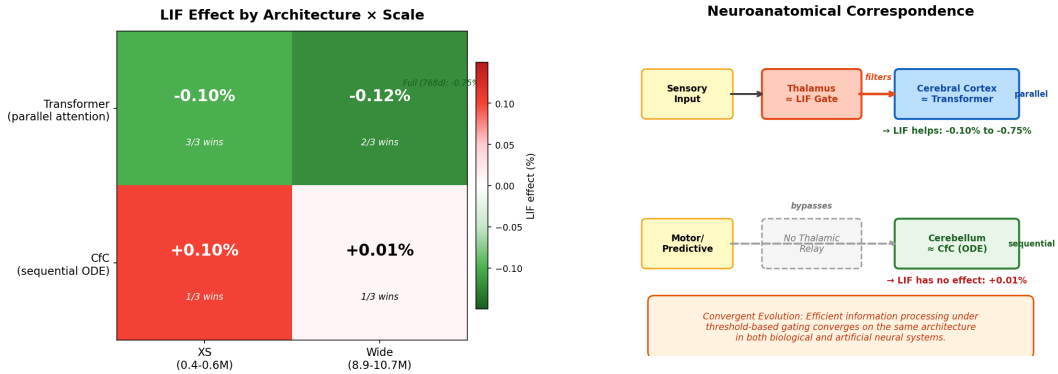

Figure 9: Cross-architecture comparison. 2×2 heatmap (Transformer/CfC $\times$ XS/Wide) showing LIF effect (%), with neuroanatomical correspondence: thalamus→cortex (LIF→Transformer, effective) vs cerebellum bypass (CfC, no effect).

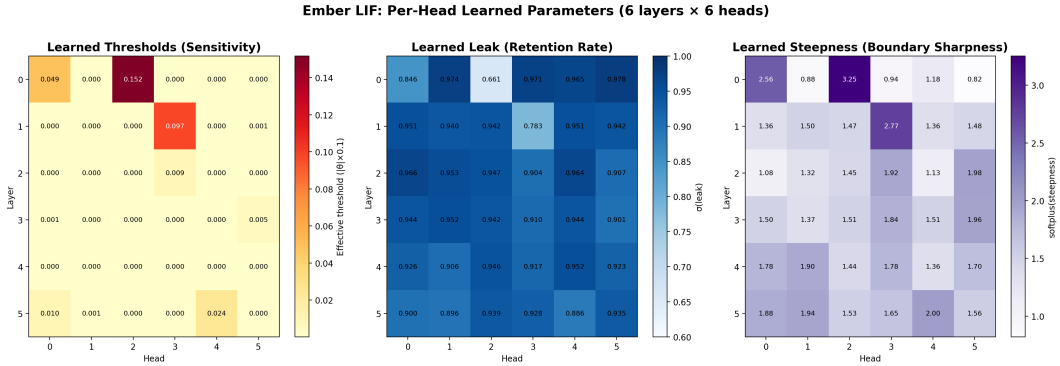

Figure 10: Per-head learned LIF parameters (wide scale, 6L/6H/384d). Heatmaps of threshold, leak, and steepness for each attention head. L0H2 emerges as an extreme outlier across all three parameters (threshold 0.152, leak 0.661, steepness 3.25), with coordinated extremity suggesting functional specialization rather than noise.

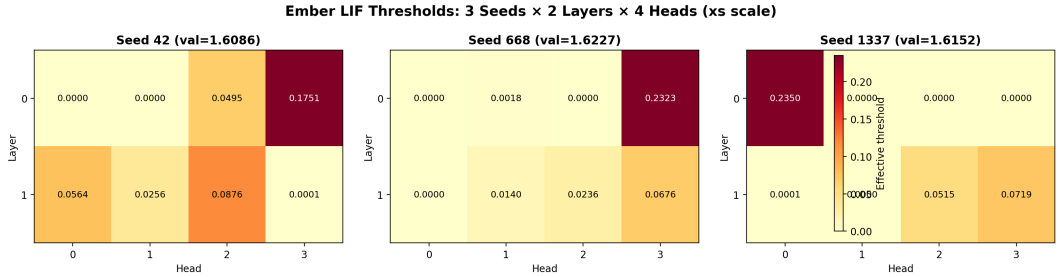

Figure 11: Cross-seed threshold differentiation (XS, 2L/4H). Learned threshold values across 3 seeds (42, 668, 1337). In these runs, each layer contains one head with a substantially higher threshold than the rest. The gatekeeper identity varies by seed (e.g., Layer 1 gatekeeper is H2 for seed 42, H3 for seeds 668 and 1337). The structural role is stable; the assignment is variable.